Errata File for Second Edition

Daniel Sorensen

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This errata file contains a list of errors, short clarifications and modifications to the manuscript's second edition.

1. Page 7, first sentence below equation (1.5): "This means that there is a 95% probability that this interval contains the true parameter θ".

   This statement is technically incorrect for the following reason. A $100\,(1-\alpha)\,\%$ confidence interval for θ is an interval defined by the lower and upper bounds $(l\,(Y)\,, u\,(Y))$, where Y represents the data, a random variable, such that

   $$\Pr\left[l\,(Y) < \theta < u\,(Y)\,|\theta\right] = 1-\alpha,$$

   where θ is the fixed, unknown parameter and α is the significance level of the test. The probability is computed with respect to the distribution of Y. Once the data are observed and $Y = y$, the expression

   $$\Pr\left[l\,(y) < \theta < u\,(y)\,|\theta\right] = 1-\alpha,$$

   does not make sense since there are no random variables involved within the square brackets. It is therefore technically incorrect to assert that the observed interval $(l\,(y)\,, u\,(y))$ contains θ with probability $(1-\alpha)$. The interval $(l\,(y)\,, u\,(y))$ either contains or does not contain θ. However, if $\alpha = 0.05$, one can be quite confident that the unobserved θ is within the observed $(l\,(y)\,, u\,(y))$ and act accordingly.

   The remaining statements in the paragraph are correct.

2. Page 170, bottom half of the page, bracket missing:
   CHANGE:

   $$\mathrm{Var}\,(\hat{a}-a) = \mathrm{E}_y(\mathrm{Var}(a|y)$$

   TO.

   $$\mathrm{Var}\,(\hat{a}-a) = \mathrm{E}_y\,(\mathrm{Var}(a|y))$$

3. Page 171, the first partial derivative below Eq. (4.62).
   CHANGE:

$$\frac{\partial \ln p\left(a,b|y,b_0,a_0,B,G,R\right)}{\partial b} = -B^{-1}(b-b_0) + XR^{-1}\left(y - Xb - Za\right)$$

   TO:

$$\frac{\partial \ln p\left(a,b|y,b_0,a_0,B,G,R\right)}{\partial b} = -B^{-1}(b-b_0) + X'R^{-1}\left(y - Xb - Za\right)$$

4. Page 172, top.
   CHANGE: "When B^{-1} implying $B = \infty$ ..."
   TO: "When and if $B^{-1} = 0$ implying $B = \infty$ ..."

5. Page 172,
   CHANGE:
   Finally, add and subtract $\hat{\theta}'\left(W'R^{-1}W + \Sigma\right)\hat{\theta}$ and rearrange to yield

$$y'R^{-1}y + \left(\theta - \widehat{\theta}\right)'\left(W'R^{-1}W + \Sigma\right)\left(\theta - \widehat{\theta}\right)' - \widehat{\theta}'\left(W'R^{-1}W + \Sigma\right)\widehat{\theta} + \theta_0'\Sigma\theta_0$$

   TO:
   Finally, add and subtract $\hat{\theta}'\left(W'R^{-1}W + \Sigma\right)\hat{\theta}$ and rearrange to yield

$$y'R^{-1}y + \left(\theta - \widehat{\theta}\right)'\left(W'R^{-1}W + \Sigma\right)\left(\theta - \widehat{\theta}\right) - \widehat{\theta}'\left(W'R^{-1}W + \Sigma\right)\widehat{\theta} + \theta_0'\Sigma\theta_0$$

6. Page 172, equation at the bottom,
   CHANGE:

$$\ln p\left(\theta|y,\theta_0,B,G,R\right) \propto -\frac{1}{2}\left(\theta - \widehat{\theta}\right)'\left(W'R^{-1}W + \Sigma\right)\left(\theta - \widehat{\theta}\right)'$$

   TO:

$$\ln p\left(\theta|y,\theta_0,B,G,R\right) \propto -\frac{1}{2}\left(\theta - \widehat{\theta}\right)'\left(W'R^{-1}W + \Sigma\right)\left(\theta - \widehat{\theta}\right)$$

7. Page 263,
   CHANGE,

$$v_\beta^* = v + 1,$$

   TO:

$$v_\beta^* = v_\beta + 1,$$

8. Page 265,
   Delete Subsection "An Alternative Parametrisation" (up to and including eqn (5.82)).

9. Page 353, Eq. (7.41), remove "comma" before "$i = 1, \ldots, m$"
   CHANGE:

$$y|\mu, \alpha, \delta, \sigma^2 \sim N\left(1\mu + \sum\nolimits_{i=1}^{m} X_i\left(\alpha_i \delta_i\right), I\sigma^2\right), \quad , i = 1, \ldots, m \tag{7.41}$$

   TO:

$$y|\mu, \alpha, \delta, \sigma^2 \sim N\left(1\mu + \sum\nolimits_{i=1}^{m} X_i\left(\alpha_i \delta_i\right), I\sigma^2\right), \quad i = 1, \ldots, m \tag{7.41}$$

10. Page 361,
    CHANGE:
    "In terms of $p-$values, a threshold of 0.05 guarantees that the expected number of false positives over the whole family of m tests is less than or equal to $0.05\,m$, far too large for modern genomewide studies."
    TO:
    "In terms of $p-$values, a threshold of 0.05 guarantees that the expected number of false positives over the whole family of m tests is $0.05\,m$, far too large for modern genomewide studies."

11. Page 361,
    CHANGE:
    "Then for the whole family of m tests, when all the m hypotheses are true nulls, the expected number of false positives is less than or equal to $m(\alpha/m) = \alpha$."
    TO:
    "Then for the whole family of m tests, when all the m hypotheses are true nulls, the expected number of false positives is equal to $m(\alpha/m) = \alpha$."

12. Page 363,
    CHANGE:
    Then the expected number of false discoveries over the m tests is

$$\mathrm{E}\left(V\right) = \mathrm{E}\left(\sum\nolimits_{i \in M_0} V_i\right) = \sum\nolimits_{i \in M_0} \mathrm{E}\left(V_i\right) \leq m_0 \alpha_B.$$

    TO:
    Then, for continuously distributed tests statistics, the expected number of false discoveries over the m tests is

$$\mathrm{E}\left(V\right) = \mathrm{E}\left(\sum\nolimits_{i \in M_0} V_i\right) = \sum\nolimits_{i \in M_0} \mathrm{E}\left(V_i\right) = m_0 \alpha_B.$$

13. Page 369,
    CHANGE:

"This is in contrast with the meaning attached to the $p-$value for $i = 19$ equal to 0.02845: in conceptual replications of the experiment where a total of m hypotheses are tested, the expected number of false discoveries among the hypotheses rejected at the cutoff 0.02845 or smaller is $0.02845 \times m$ or smaller."
TO:
"This is in contrast with the meaning attached to the $p-$value for $i = 19$ equal to 0.02845: in conceptual replications of the experiment where a total of m hypotheses are tested of which m_0 are true nulls, for continuously distributed test statistics, the expected number of false discoveries is $0.02845\, m_0$."

14. Page 373, paragraph above Eq. (8.23), defines the number of false discoveries $V(A)$, given that k hypotheses are rejected, as binomially distributed $Bi(k, \Pr(H = 0|z \in A)$. Another definition can be found on page 376, paragraph above Eq. (8.33), concerning the expected number of false discoveries. Both definitions may benefit from a little explanation that is provided below.

Background

Let the binary random variable $V_i = 1$ if hypothesis i is rejected and if hypothesis i is a true null; $V_i = 0$, otherwise. Using the notation on page 363, we can write $V_i = I(H_i = 0, R_i = 1)$, where $R_i = 1$ if $z_i \in A$ and z_i is the observed value of the scalar random variable Z_i. Treating the hypotheses as random, it follows that V_i is a Bernoulli random variable with probability $\Pr(H_i = 0, Z_i \in A)$; that is $V_i \sim Br(\Pr(H_i = 0, Z_i \in A)$.

$$V_i \sim Br(\Pr(H_i = 0, Z_i \in A).$$

- **Page 373**. Consider the binary random variable $V_i|R_i = 1$ that takes the value 1 if $H_i = 0$ and 0 otherwise. Then $[V_i|R_i = 1] \sim Br(\Pr(H_i = 0|Z_i \in A)$, a Bernoulli distributed random variable. If in a series of independent trials the total number of rejections is $R = k$, among these k rejections, the total number of false discoveries is binomially distributed:

  $$[V|R = k] \sim Bi(k, \Pr(H = 0|Z \in A).$$

  The expected number of false discoveries among the k rejections is

  $$\mathrm{E}(V|R = k) = k\ \Pr(H = 0|Z \in A).$$

  The subscript i has been omitted because $\Pr(H_i = 0|Z_i \in A)$ is the same for each i.

- **Page 376**. Consider an experiment where m independent hypotheses are tested. Given the definition of V_i and the independence assumption, the total number of false discoveries, $V = \sum_{i=1}^{m} V_i$, is then binomially distributed

  $$V \sim Bi(m, \Pr(H = 0, Z \in A)),$$

with expected value

$$\mathrm{E}(V) = m\ \Pr(H = 0, Z \in A)) = m\ \Pr(Z \in A | H = 0)\pi_0,$$

where $\pi_0 = \Pr(H = 0)$, the prior probability of a true null.

Note that: (i) hypotheses H_i are treated as random (with a common prior π_0), (ii) random vectors (H_i, Z_i), $i = 1, 2, \ldots, m$, are independently distributed, (iii) the rejection region A is data-independent and fixed in advance.